

Hostel Management System

Project Report

1. Cover Page

Hostel Management System

College Project Report

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Course: BTech AI ML CSE

College: VIT Bhopal

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2. Introduction

The Hostel Management System is a simple Python-based application designed to help hostel administrators efficiently manage student accommodation details, including registration, room allocation, and fee status. This system reduces manual errors and streamlines administrative tasks.

3. Problem Statement

Managing student records manually in a hostel environment is time-consuming and error-prone. This project aims to automate the record-keeping process by providing a system that stores student information securely and allows easy addition, viewing, and removal of student data.

4. Functional Requirements

- Add new student records with registration number, name, branch, room number, and fee status.
- View the list of all students currently registered.
- Remove a student record by registration number.
- View statistics about total students, fee payments completed, and pending fees.
- Persistent storage of data in CSV files.

5. Non-functional Requirements

- The system should be easy to use via a command-line interface.
- Data storage must be persistent across sessions using CSV files.
- The system must handle errors gracefully (e.g., duplicate entries, missing records).
- The software should be lightweight and require no additional dependencies beyond standard Python libraries.

6. System Architecture

The system is structured around two main components:

- **Student Class:** Represents individual student data.
- **HostelManager Class:** Manages student records, handles file operations, and provides the user interface through CLI menus.

The application interacts with the file system for persistent CSV storage and provides a menu-driven interface for user interaction.

7. Design Diagrams

Use Case Diagram

- Actors: Hostel Administrator
- Use Cases: Add Student, View Students, Remove Student, View Statistics, Exit

Workflow Diagram

- Start → Display Menu → User Chooses Option → Perform Action → Loop Back or Exit

Sequence Diagram

- User → HostelManager: select option
- HostelManager → Student: add/view/remove student
- HostelManager → CSV File: read/write data

Class Diagram

- **Classes:**
 - **Student:** Attributes (reg, name, branch, room, feestat)
 - **HostelManager:** Methods (addstudent, viewall, removestudent, stats, save, load)

ER Diagram

Not applicable (CSV file-based storage without relational database).

8. Design Decisions & Rationale

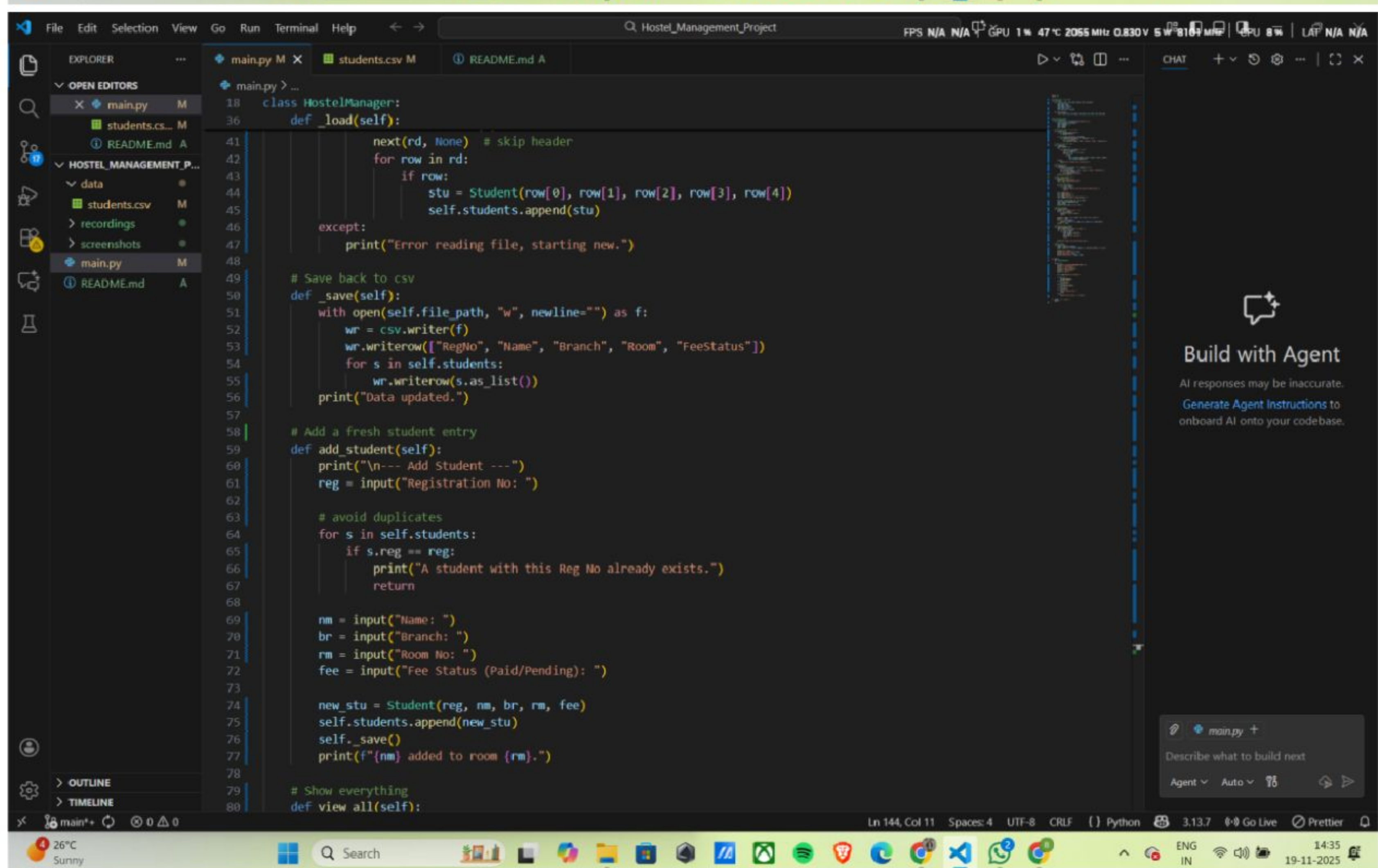
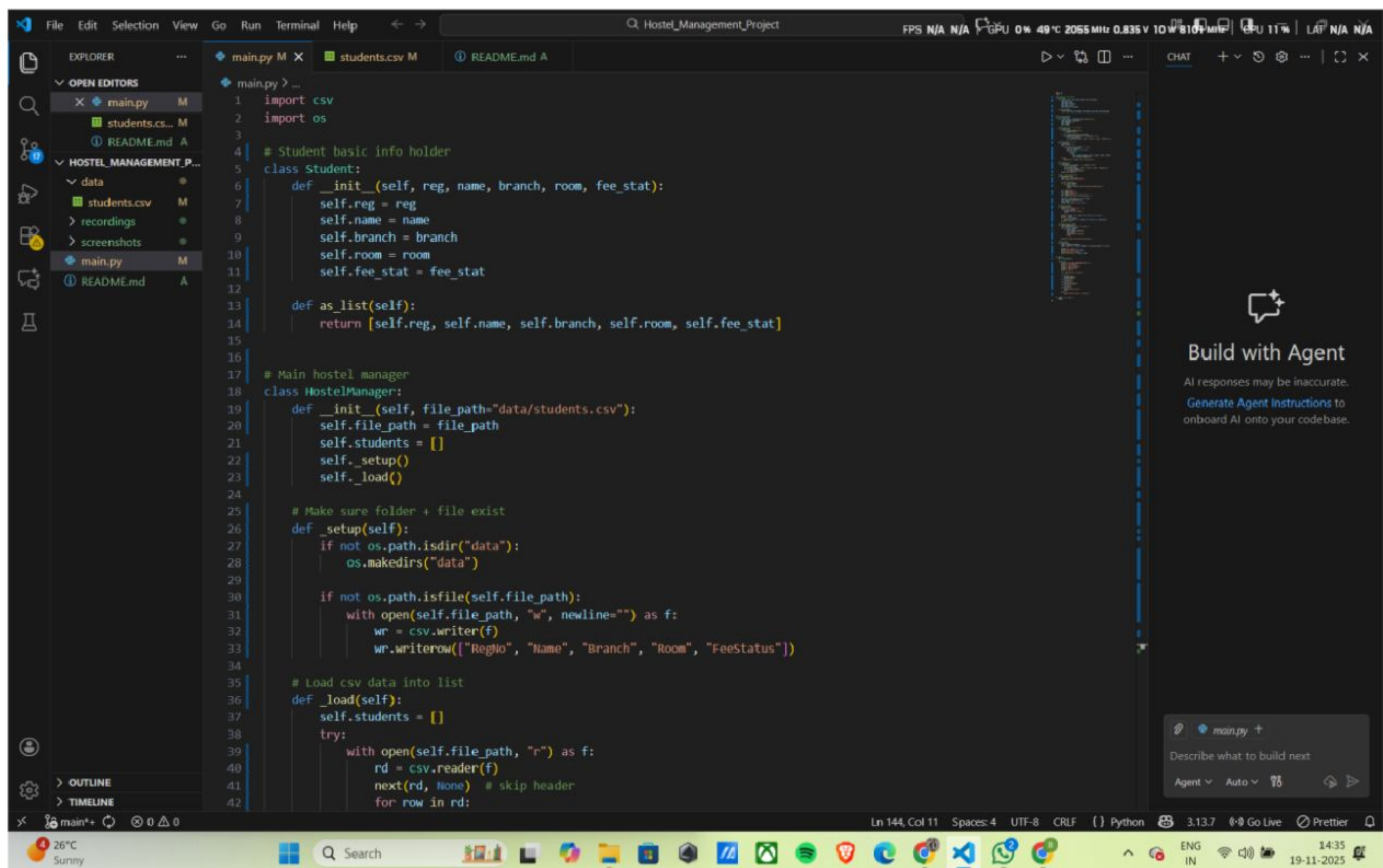
- **CSV File Storage:** Chosen for simplicity and ease of use without requiring a database setup.
 - **Command-line Interface:** Enables wide compatibility without GUI dependencies.
 - **Data Validation:** Ensures no duplicate student registration numbers are allowed.
 - **Modular Design:** Separation of concerns between data model (Student class) and management logic (HostelManager class).
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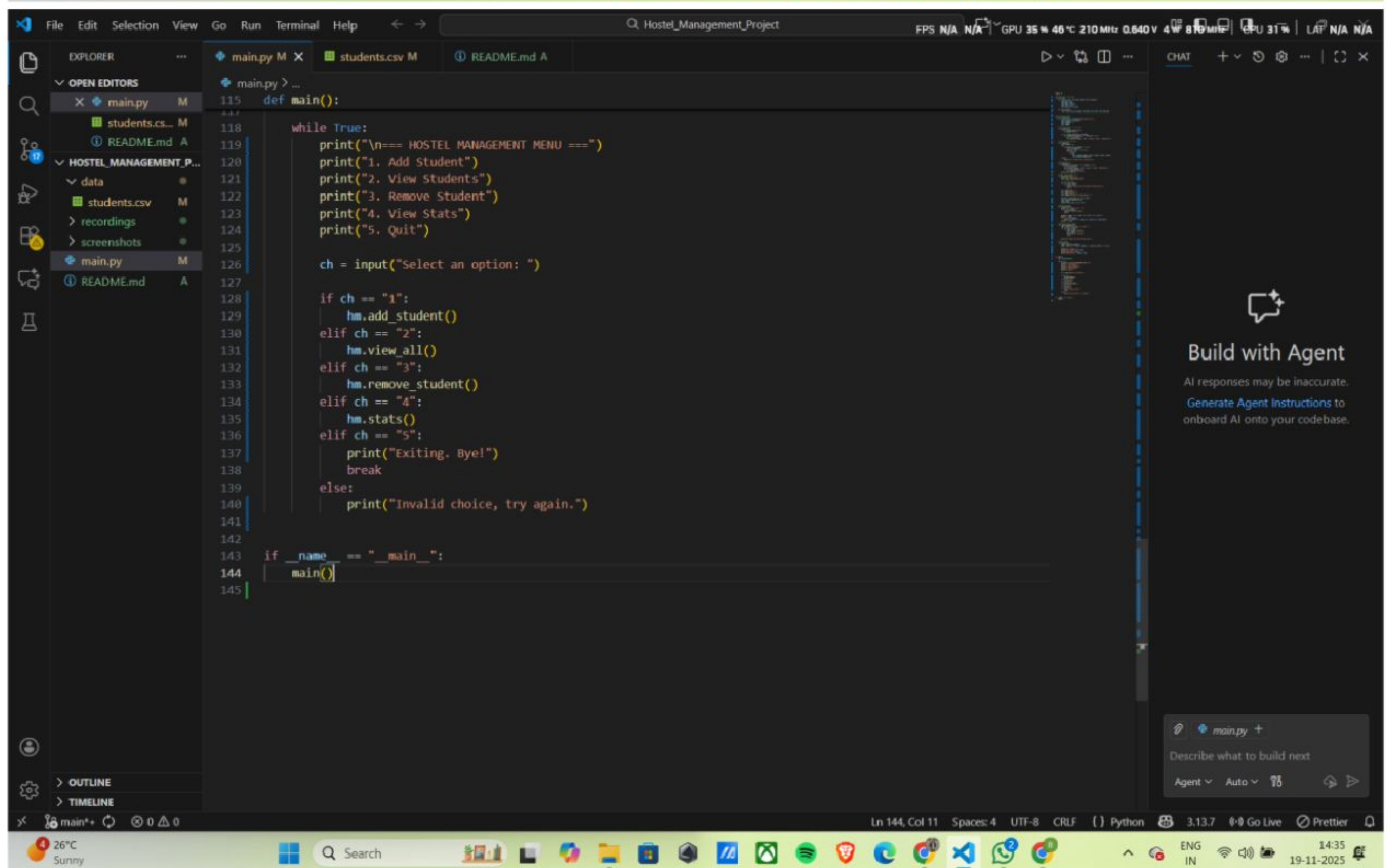
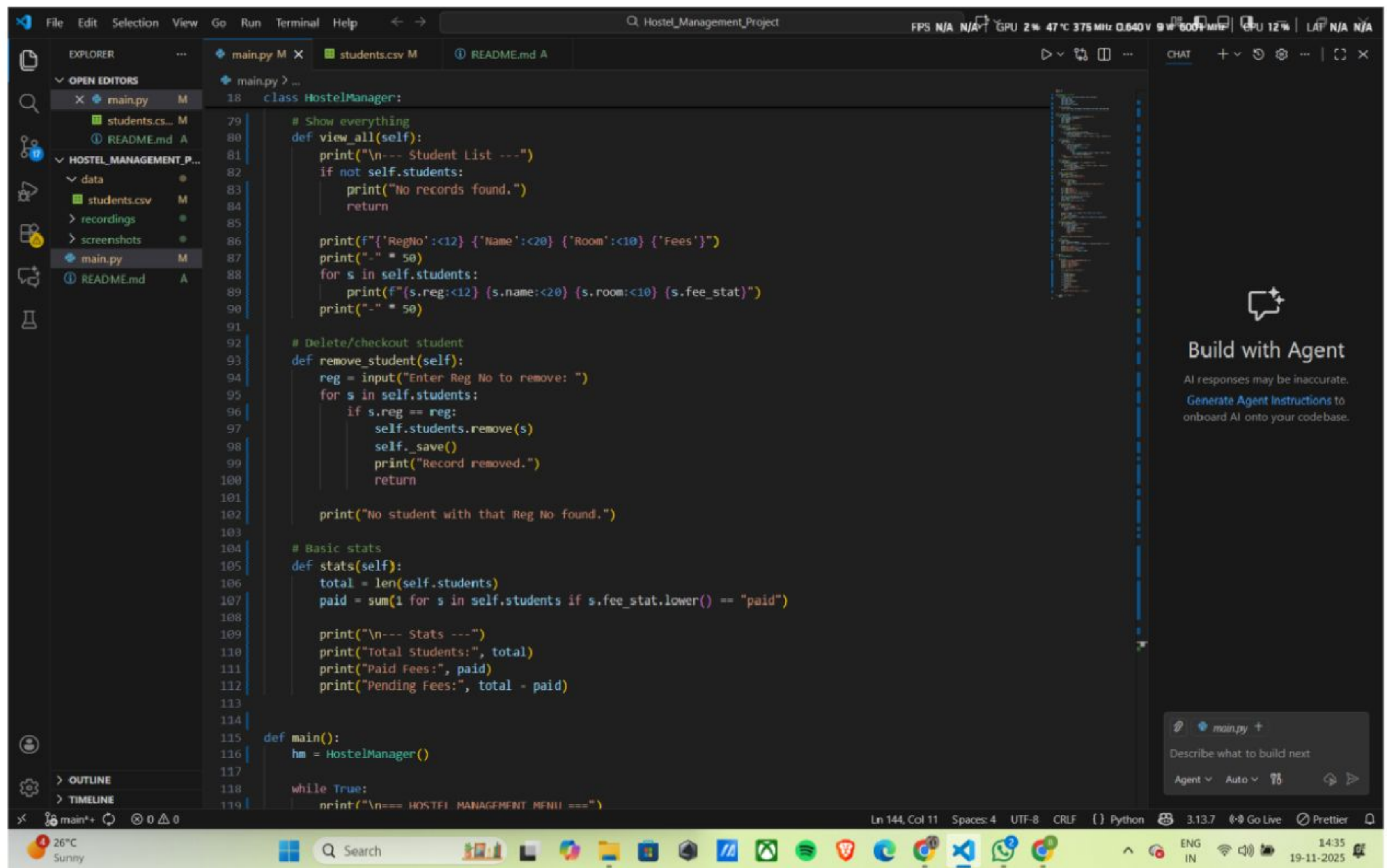
9. Implementation Details

The project is implemented in Python using standard libraries (csv and os).

- The Student class encapsulates student attributes and provides a method to convert data into list form for CSV writing.
 - The HostelManager class manages CRUD operations, handles file reading/writing, and provides the user menu for interaction.
 - Data persistence is handled by reading from and writing to data/students.csv.
 - The program ensures the data directory and CSV file exist, creating them if necessary.
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10. Screenshots / Results





File Edit Selection View Go Run Terminal Help

Hostel_Management_Project

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EXPLORER

main.py
students.csv
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PS C:\Users\ASUS\Desktop\Hostel_Management_Project> & C:/Python313/python.exe c:/Users/ASUS/Desktop/Hostel_Management_Project/main.py

==== HOSTEL MANAGEMENT MENU ===
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 2

--- Student List ---
RegNo Name Room Fees

25BCE10587 Devansh Gaur A416 Paid
25BCE10588 Bhavishya Gupta A416 Paid
25BAT20251 Aryan Gupta A411 Paid
djhjdbj asdvh paid
25BCE10590 Aadarsh Aggarwal A712 Pending
25BCE10585 Abhinav Sahu A124 Paid

==== HOSTEL MANAGEMENT MENU ===
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 1

--- Add Student ---
Registration No: 25BCE10466
Name: Raghav Kesarwani
Branch: CSE Core
Room No: A671
Fee Status (Paid/Pending): Paid
Data updated.
Raghav Kesarwani added to room A671.

==== HOSTEL MANAGEMENT MENU ===
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 2

Python
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Build with Agent

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Generate Agent Instructions to onboard AI onto your codebase.

main.py +
Describe what to build next
Agent Auto

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3. Remove Student
4. View Stats
5. Quit
Select an option: 2

--- Student List ---
RegNo Name Room Fees

25BCE10587 Devansh Gaur A416 Paid
25BCE10588 Bhavishya Gupta A416 Paid
25BAT20251 Aryan Gupta A411 Paid
djhjdbj asdvh paid
25BCE10590 Aadarsh Aggarwal A712 Pending
25BCE10585 Abhinav Sahu A124 Paid
25BCE10466 Raghav Kesarwani A671 Paid

==== HOSTEL MANAGEMENT MENU ===
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 3
Enter Reg No to remove: djhjdbj
Data updated.
Record removed.

==== HOSTEL MANAGEMENT MENU ===
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 2

--- Student List ---
RegNo Name Room Fees

25BCE10587 Devansh Gaur A416 Paid
25BCE10588 Bhavishya Gupta A416 Paid
25BAT20251 Aryan Gupta A411 Paid
25BCE10590 Aadarsh Aggarwal A712 Pending
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Select an option: 3
Enter Reg No to remove: djhjdbj
Data updated.
Record removed.

==== HOSTEL MANAGEMENT MENU ====
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 2

--- Student List ---

RegNo	Name	Room	Fees
25BCE10587	Devansh Gaur	A416	Paid
25BCE10588	Bhavishya Gupta	A416	Paid
25BAI20251	Aryan Gupta	A411	Paid
25BCE10590	Aadarsh Aggarwal	A712	Pending
25BCE10585	Abhinav Sahu	A124	Paid
25BCE10466	Raghav Kesarwani	A671	Paid

==== HOSTEL MANAGEMENT MENU ====
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 4

--- Stats ---
Total Students: 6
Paid Fees: 5
Pending Fees: 1

==== HOSTEL MANAGEMENT MENU ====
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 5
Exiting. Bye!
PS C:\Users\VASUS\Desktop\Hostel_Management_Project>

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Screenshot (9)... A
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data > students.csv > data
1 RegNo,Name,Branch,Room,FeeStatus
2 25BCE10587,Devansh Gaur,CSE Core,A416,Paid
3 25BCE10588,Bhavishya Gupta,CSE Core,A416,Paid
4 25BAI20251,Aryan Gupta,CSE AI/ML,A411,Paid
5 25BCE10590,Aadarsh Aggarwal,CSE Core,A712,Pending
6 25BCE10585,Abhinav Sahu,CSE Core,A124,Paid
7 25BCE10466,Raghav Kesarwani,CSE Core,A671,Paid
8

--- Stats ---
Total Students: 6
Paid Fees: 5
Pending Fees: 1

==== HOSTEL MANAGEMENT MENU ====
1. Add Student
2. View Students
3. Remove Student
4. View Stats
5. Quit
Select an option: 5
Exiting. Bye!
PS C:\Users\VASUS\Desktop\Hostel_Management_Project>

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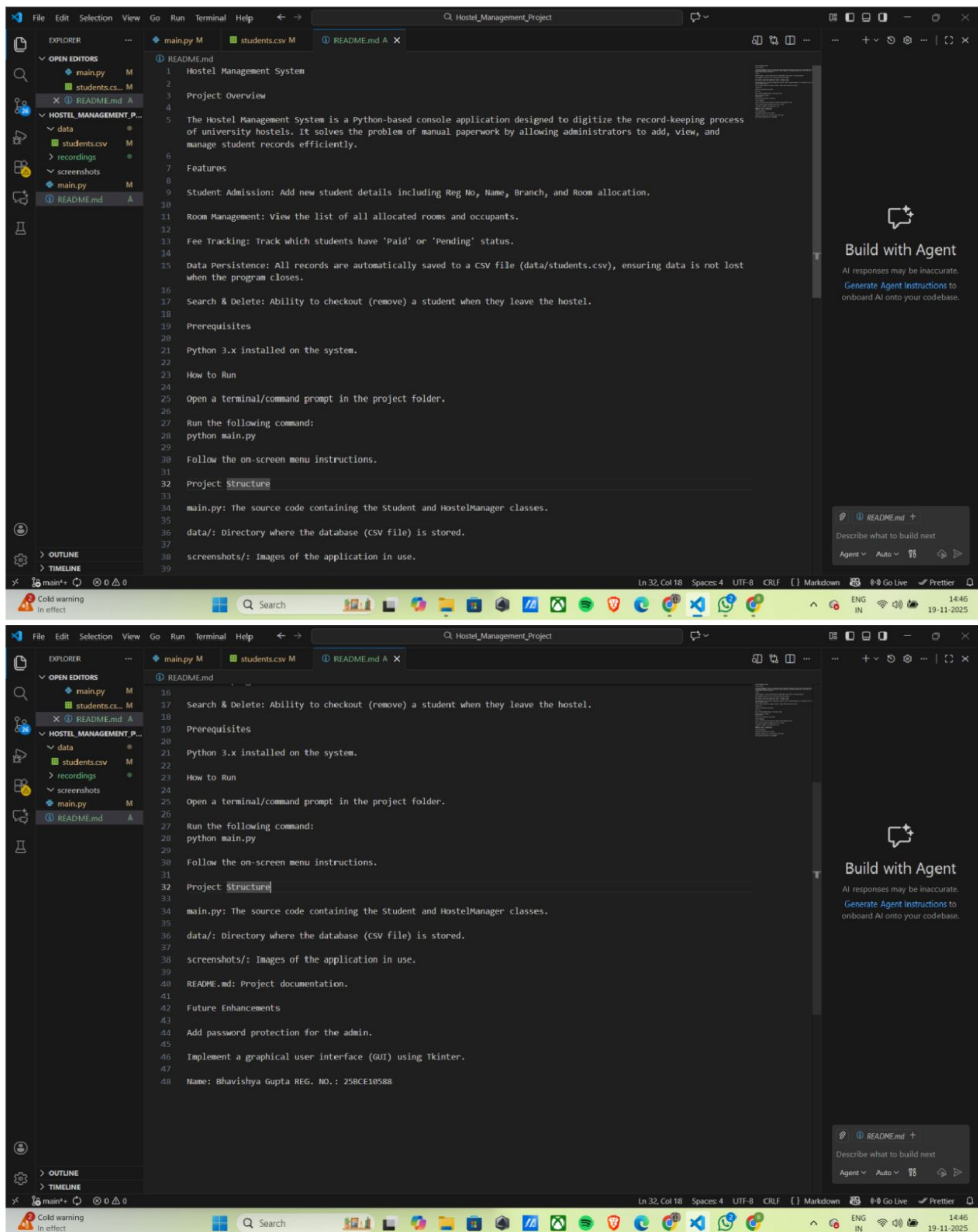
Describe what to build next

Agent Auto

main+ 0 0 0

Col 1: RegNo Ln 8, Col 1 Spaces: 4 UTF-8 CRLF CSV 3.13.7 Go Live Prettier

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11. Testing Approach

- Manual testing by running the program and interacting with each menu option.
- Tested for duplicate student entries to ensure the system prevents duplicates.
- Tested removal of existing and non-existing records to check error handling.
- Verified data persistence by restarting the program and checking data integrity.

12. Challenges Faced

- Handling file read/write exceptions and ensuring data consistency.
 - Managing user input validation in a command-line environment.
 - Structuring the code to maintain modularity while keeping it simple.
-

13. Learnings & Key Takeaways

- Gained practical experience in file handling with Python.
 - Understood the importance of modular code design for maintainability.
 - Improved skills in CLI-based user interaction design.
 - Learned about data validation and error handling in simple systems.
-

14. Future Enhancements

- Develop a graphical user interface (GUI) for better user experience.
 - Integrate a database system (e.g., SQLite) for more scalable data management.
 - Add features like fee reminders and automated reports.
 - Implement user authentication for secure access.
-

15. References

- Python Official Documentation: <https://docs.python.org/3/>
- CSV Module Documentation: <https://docs.python.org/3/library/csv.html>
- Various online tutorials on Python file handling and CLI applications.